CO3-AT2-Q1:

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Railway Reservation Search System Using Binary Search

1. Problem Statement

A railway reservation system contains a large number of train schedules and booking records. Searching a particular train number from thousands of records manually is time-consuming. This case study implements **Binary Search** to efficiently retrieve train information from a sorted database.

2. Objective

- To search train records quickly using Binary Search.
- To apply the Divide and Conquer approach.
- To analyze the efficiency and performance of the algorithm.
- To understand the importance of sorted indexing in database searching.

3. Problem Understanding

The railway database stores train numbers in sorted order. The system needs to find whether a particular train number exists or not.

Input:

- Number of train records
- Sorted train numbers
- Train number to search

Output:

- Position of the train record if found
- Message if record is not available

4. Algorithm Used – Binary Search

Binary Search follows the Divide and Conquer technique.

Steps:

1. Take the middle element of the sorted list.
2. Compare the middle value with the required train number.
3. If both are equal, the record is found.
4. If the search value is smaller, search the left half.
5. If the search value is larger, search the right half.
6. Repeat until the element is found or the search space becomes empty.

5. Requirements

Sorted Indexing

Binary Search requires data to be arranged in sorted order. Train numbers are stored in ascending order to improve searching speed.

Speed

Binary Search reduces the search area by half in every step, making it faster for large railway databases.

Accuracy

The algorithm compares exact values and provides reliable search results.

6. Time Complexity Analysis

Case	Complexity
Best Case	O(1)
Average Case	O(log n)
Worst Case	O(log n)

Space Complexity

O(1)

7. Result

The Binary Search algorithm successfully retrieves train records from a sorted railway reservation database with high efficiency.